

SOLUTIONS Exam 2 (November 9/13)

QTM 220

Problem 1

Suppose we have a sample of $Y_1 \dots Y_n$ and $X_1 \dots X_n$ stored in an **R** dataframe 'df' with n rows. X can take the values of 1,2,3,4 or 5. We run this code.

```
B=10000
theta<- rep(NA,B)
for( b in 1:B ){
  bootind = sample(1:n,size=n,replace=T)
  dfboot = df[bootind,]
  theta[b] = (1/4)*(mean(dfboot$y[dfboot$x==5]) - mean(dfboot$y[dfboot$x==1]))
}
```

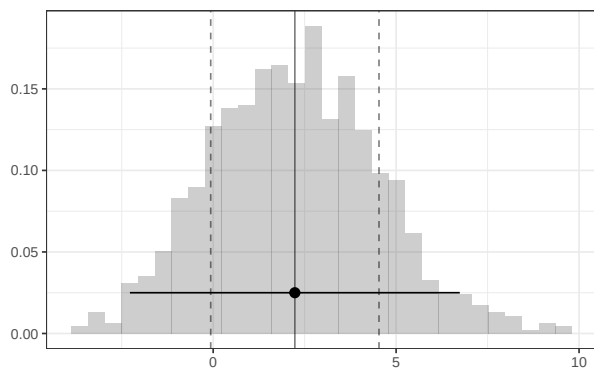


Figure 1: Histogram of the vector Y's elements

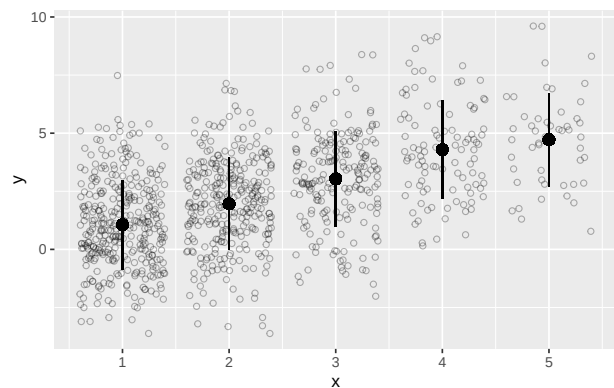


Figure 2: Jittered Scatterplot of X and Y

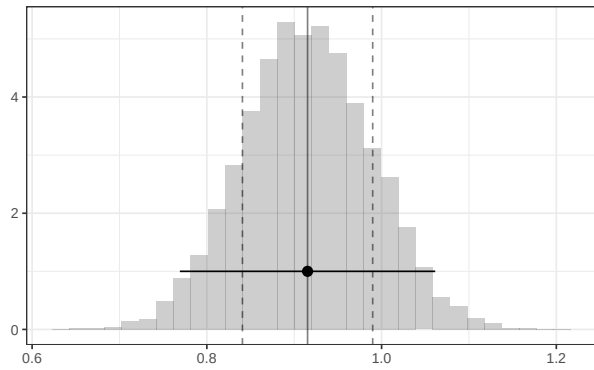


Figure 3: Histogram of the theta's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

The sample mean (unconditioned). It's indicated by the solid vertical line on the sample histogram.

Part b

What does this code calculate? State it in words and mark it with a 'b', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

The sample standard deviation. It's the distance between the solid vertical and either dashed vertical line on the sample histogram.

Part c

What does this code calculate? State it in words and mark it with a ‘c’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y[X==1])
```

The conditional sample mean of Y among X=1. It's indicated by the black dot on the scatterplot where X=1.

Part d

What does this code calculate? State it in words and mark it with a ‘d’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(theta)
```

The bootstrap sampling distribution's standard deviation for the average increment estimator. It's the distance between the solid vertical and either dashed vertical line on the sampling distribution histogram.

Part e

What does this code calculate? State it in words and mark it with a ‘e’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
(1/4)(mean(Y[X=5]) - mean(Y[X=1]))  
- 1.96* sqrt((1/4)^2*var(Y[X=5])/length(Y[X=5]) + (-1/4)^2*var(Y[X=1])/length(Y[X=1]))  
  
(1/4)(mean(Y[X=5]) - mean(Y[X=1]))  
+ 1.96* sqrt((1/4)^2*var(Y[X=5])/length(Y[X=5]) + (-1/4)^2*var(Y[X=1])/length(Y[X=1]))
```

A 95% confidence interval — based on normal approximation — for the average incremental difference in Y from X=1 to X=5 in the population from which our data was drawn. This is indicated by

the horizontal segment (with a dot in the middle) on the sampling distribution histogram (theta).

Part f

How could you calculate roughly the same thing as in part e using 'sd(theta)' replacing

$$\sqrt{((1/4)^2 \text{var}(Y[X=5]) / \text{length}(Y[X=5]) + (-1/4)^2 \text{var}(Y[X=1]) / \text{length}(Y[X=1]))}$$

We can use the standard deviation of the bootstrap sampling distribution as our estimate of standard error.

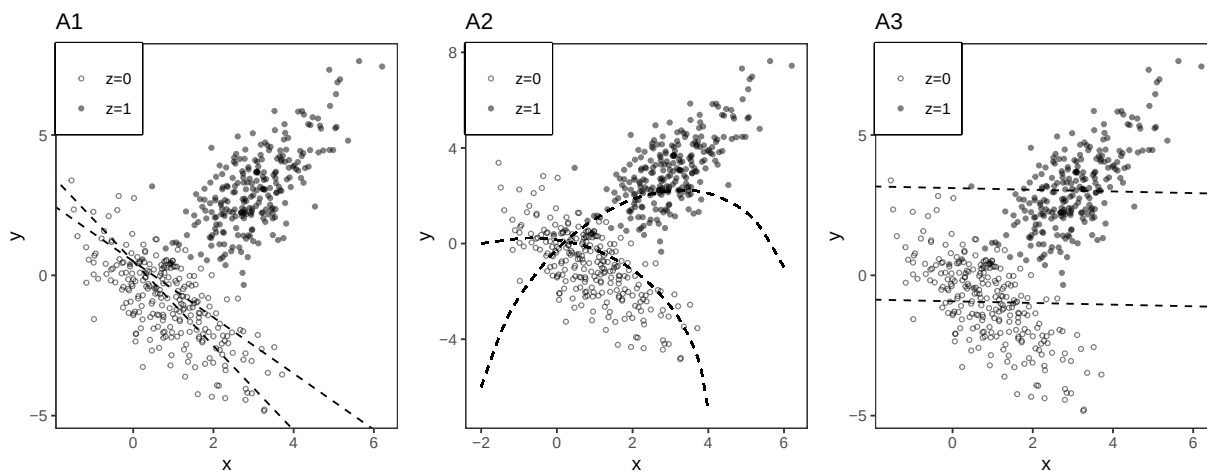
$$\begin{aligned} (1/4) (\text{mean}(Y[X=5]) - \text{mean}(Y[X=1])) - 1.96 * \text{sd}(\text{theta}) \\ (1/4) (\text{mean}(Y[X=5]) - \text{mean}(Y[X=1])) + 1.96 * \text{sd}(\text{theta}) \end{aligned}$$

Problem 2

In the following parts, choose the figure with the appropriate regression line(s).

Part A

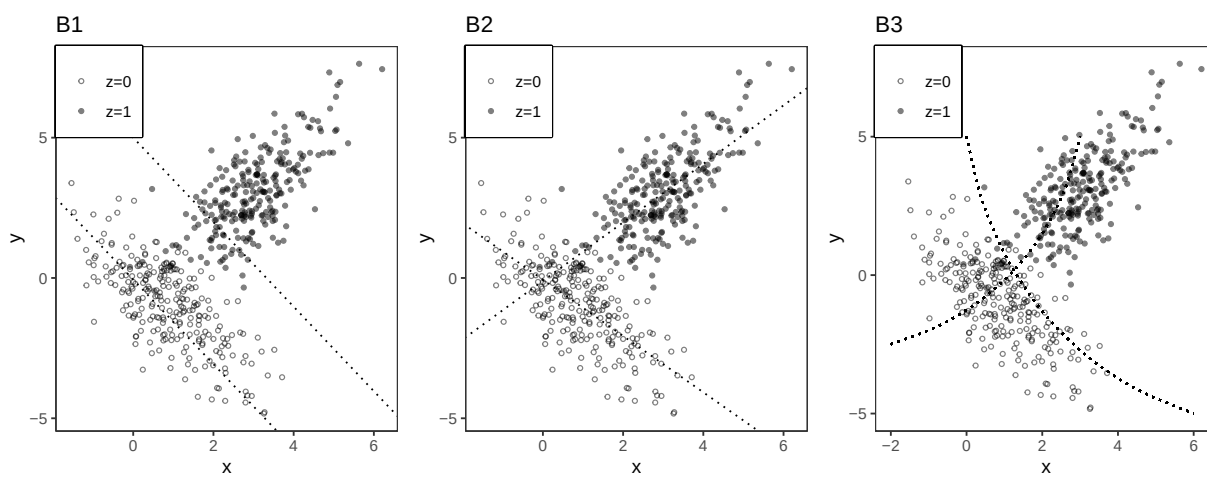
Choose the figure with the two dashed lines associated with the additive multiple regression ($y \sim x + z$).



A3

Part B

Choose the figure with the two dotted lines associated with the interactive multiple regression ($y \sim x * z$).



B2

Problem 3

Part A

We have collected y , x , and a binary variable z . The scatter plots below show our data.

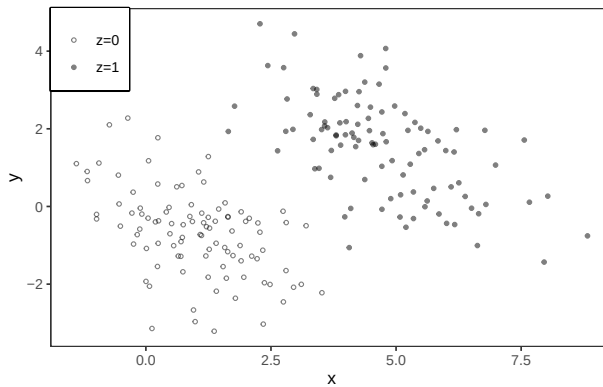


Figure 4: Scatterplot 1

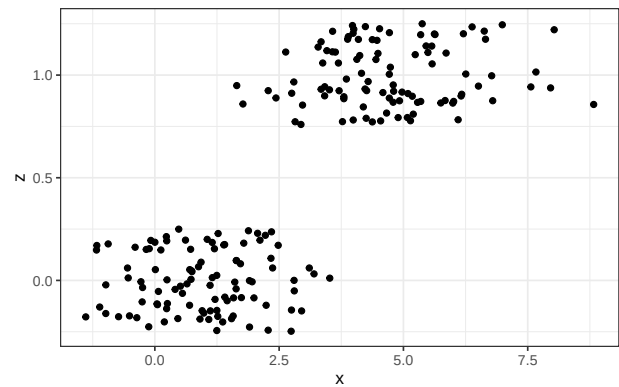


Figure 5: Scatterplot 2 (jittered)

Part A

Do you expect the coefficient on x from the simple regression of y on x ($lm(y \sim x)$) to be: (choose one, no explanation needed)

- Positive
- Negative
- Zero

Part B

We then run the regression $lm(y \sim x + z)$ which gives us the following output:

Call:

```
lm(formula = y ~ x + z, data = dfp3)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.9982	-0.6313	-0.0448	0.7168	3.2627

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.07683	0.11938	-0.644	0.521
x	-0.55172	0.05963	-9.253	<2e-16 ***
z	4.21009	0.26966	15.613	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.05 on 197 degrees of freedom

Multiple R-squared: 0.5961, Adjusted R-squared: 0.592

F-statistic: 145.4 on 2 and 197 DF, p-value: < 2.2e-16

And we run the regression $lm(z \sim x)$ which gives us the following output:

Call:

`lm(formula = z ~ x, data = dfp3)`

Residuals:

	Min	1Q	Median	3Q	Max
	-0.62555	-0.20421	-0.00143	0.20566	0.71973

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.023784	0.031416	-0.757	0.45
x	0.184544	0.008657	21.317	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2768 on 198 degrees of freedom

Multiple R-squared: 0.6965, Adjusted R-squared: 0.695

F-statistic: 454.4 on 1 and 198 DF, p-value: < 2.2e-16

How would you use these outputs to calculate the coefficient on x from the simple regression $lm(y \sim x)$? Explain your steps and define any notation you use. Plug in the estimates from your output (you can leave it as an equation).

Let the simple regression of y on x be $y = \beta_0^S + \beta_1^S x$

Let the multiple regression of y on x and z be $y = \beta_0 + \beta_1 x + \beta_2 z$

Let the regression of z on x be $z = \gamma_0 + \gamma_1 x$ where $\gamma_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})z_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$

From our OVB formula, we know $\beta_1^S = \beta_1 + \beta_2 * \gamma_1$

The regression outputs indicate that $\beta_1 = -0.5517$, $\beta_2 = 4.21$, and $\gamma_1 = 0.1845$

$$\beta_1^S = -0.5517 + (4.21) * (0.1845) = 0.225$$

Part C

Suppose we want β_1 from $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i$ but we run the regression $lm(y \sim x)$ without z or w . We have de-meaned our variables x_i , z_i , and w_i such that $\bar{x} = \sum_{i=1}^n x_i = 0$, $\bar{z} = \sum_{i=1}^n z_i = 0$, and $\bar{w} = \sum_{i=1}^n w_i = 0$. Under what condition(s) will the coefficient on x from $lm(y \sim x)$, which is $\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$, be equal to β_1 ? Derive what your coefficient on x will be in terms of β s and justify your steps.

The coefficient on x from $lm(y \sim x)$ is

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

Plug in y_i defined by the multivariate regression: $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i$

$$\frac{\sum_{i=1}^n x_i (\beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i)}{\sum_{i=1}^n x_i^2}$$

Distribute

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_3 w_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2}$$

Let's work through each term.

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 * 0}{\sum_{i=1}^n x_i^2} = 0$$

as x is demeaned, $\sum x_i = 0$

$$\frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_1 \sum_{i=1}^n x_i x_i}{\sum_{i=1}^n x_i^2} = \beta_1$$

$$\frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\sum_{i=1}^n x_i \beta_3 w_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_3 \sum_{i=1}^n x_i w_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2} = \frac{0}{\sum_{i=1}^n x_i^2} = 0$$

we know that $\sum_{i=1}^n x_i e_i = 0$ from the first order conditions on our least squares estimator (we chose the intercept and coefficient(s) that minimize the sum of squared residuals)

Thus we have

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \beta_1 + \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2} + \frac{\beta_3 \sum_{i=1}^n x_i w_i}{\sum_{i=1}^n x_i^2}$$

The coefficient on x from the simple regression of y on x will be equal to β_1 if: (1) β_2 or the coefficient on x from the simple regression from z on x is 0 and (2) β_3 or the coefficient on x from the simple regression of w on x is 0. If either of these terms is non-zero, our simple regression will not give us the same result as our regression that includes both x , z , and w . If x is randomized (like a random treatment assignment), it should not be correlated with z or w , so our short regression coefficient should give us β_1 .

Problem 4

Consider the following R code:

```
1. lm(y ~ residuals(lm(z~x)))
2. lm(y ~ residuals(lm(x~z)))
3. lm(residuals(lm(y~z)) ~ x)
4. lm(residuals(lm(y~x)) ~ z)
5. lm(residuals(lm(y~x)) ~ residuals(lm(z~x)))
6. lm(residuals(lm(y~x)) ~ residuals(lm(x~z)))
7. lm(residuals(lm(y~z)) ~ residuals(lm(x~z)))
8. lm(residuals(lm(y~z)) ~ residuals(lm(z~x)))
```

- a) Choose one of the above that will produce the coefficient on x from the multiple regression of $y \sim x + z$

2 or 7

- b) BONUS: Choose one of the above that will produce the coefficient on z from the multiple regression of $y \sim x + z$

5 or 1

- c) Choose one of the above that will produce the same residuals as the residuals from the multiple regression of $y \sim x + z$

7 or 5